

Gayatri Chakilam

Data Engineer

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SUMMARY

- Results-driven Data Engineer with 3+ years of hands-on experience in designing and implementing scalable data pipelines, ETL processes, and analytics solutions using modern cloud technologies and distributed computing frameworks
- Proficient in Python programming with extensive knowledge of data manipulation libraries including Pandas, NumPy, and Scikit-learn, successfully processing and analyzing datasets exceeding 100GB
- Strong foundation in cloud platforms (AWS, Azure, GCP) with practical experience in implementing serverless architectures, data lakes, and enterprise-scale data warehousing solutions
- Demonstrated expertise in SQL and NoSQL databases, having worked with both traditional and modern data storage solutions while maintaining high performance and data integrity
- Advanced skills in data visualization and business intelligence tools, creating impactful dashboards that drive data-informed decision making and improve stakeholder engagement
- Extensive experience with big data technologies including Apache Spark and distributed computing frameworks, processing large-scale datasets efficiently
- Proven ability to collaborate effectively with cross-functional teams and communicate complex technical concepts to non-technical stakeholders, resulting in successful project deliveries

EDUCATION

Masters in Computer Science | Wright State University, Dayton, OH – USA
Bachelor's degree in Computer Science | CMR Institute of Technology, Hyderabad – INDIA

SKILLS

- **Programming:** Python (NumPy, Pandas, Matplotlib, SciPy), PySpark, SQL, Java
- **Cloud & ETL:** AWS (Lambda, Glue, S3, Athena, Redshift, QuickSight, Step Functions, IAM), Azure, GCP
- **Data Warehousing:** Snowflake, Amazon Redshift, Azure Synapse Analytics
- **Databases:** MySQL, PostgreSQL, MongoDB, Cassandra, DynamoDB
- **Big Data:** Apache Spark, Hadoop, Amazon EMR, Databricks
- **Data Visualization:** Tableau, Power BI, Amazon QuickSight, Seaborn, Plotly
- **Machine Learning:** Scikit-learn, TensorFlow, Keras, Natural Language Processing
- **BI & Analytics:** SSIS, SSRS, SSAS, Looker, Amazon QuickSight
- **Other Tools:** Git, Docker, Kubernetes, Airflow, Jupyter Notebook
- **Soft Skills:** Data Storytelling, Problem-Solving, Communication, Agile Methodologies

EXPERIENCE

BNY, Ohio | Data Engineer

September 2023 – Current

- Designed and implemented serverless ETL pipelines using AWS Glue, Lambda functions, and Step Functions, processing 200+ GB of daily data across multiple data sources while maintaining 99.9% job success rate
- Developed and optimized AWS Glue jobs using PySpark for data transformation and loading, reducing processing time by 45% and implementing robust error handling and monitoring mechanisms
- Engineered and maintained EMR clusters for large-scale data processing, implementing automated scaling policies and Spark optimization techniques that reduced processing costs by 40% while handling 500+ GB daily workloads
- Engineered data lake solutions using AWS S3 and Athena, creating efficient partitioning strategies and implementing column-level encryption, resulting in 60% faster query performance and enhanced data security
- Created and optimized Amazon Redshift tables and materialized views, implementing proper distribution and sort keys that improved query performance by 40% and reduced storage costs by 30%
- Created automated data validation frameworks using AWS Lambda and Python (Pandas, NumPy) to perform quality checks on incoming data, reducing data anomalies by 70% and ensuring data consistency across pipelines
- Implemented streaming data pipelines using Amazon Kinesis and Lambda functions to process real-time financial transactions, handling 10,000+ events per second with sub-second latency
- Developed and maintained Snowflake data warehouse, including design of multi-layer architecture, implementation of proper data modeling practices, and creation of efficient ELT processes that improved data accessibility by 50%
- Built comprehensive monitoring dashboards using Amazon QuickSight and Grafana, providing real-time visibility into ETL job performance and data quality metrics, reducing incident response time by 65%

DataOne, India | Data Analyst

February 2021 – July 2022

- Developed and maintained large-scale data analysis workflows using Python (Pandas, NumPy) and advanced statistical methods, processing multiple terabytes of data to identify patterns and trends that increased operational efficiency by 25% and reduced processing costs by \$50K annually
- Architected and implemented Snowflake data warehousing solutions, including design of multi-layer data architecture, implementation of proper data modeling practices, and optimization of warehouse performance, resulting in 40% improvement in query performance and 30% reduction in storage costs
- Engineered automated data pipeline using Azure Data Factory and Python, implementing parallel processing and incremental loading patterns that reduced data processing time from 4 hours to 30 minutes while maintaining data consistency and reliability
- Created dynamic and interactive visualization dashboards using Tableau and Power BI, incorporating advanced analytics and drill-down capabilities that increased stakeholder engagement with data insights by 45% and improved data-driven decision making across departments
- Designed and implemented comprehensive data quality monitoring framework using custom Python scripts and SQL validation rules, reducing data errors by 60% and implementing automated notification systems for data quality issues
- Developed RESTful APIs using Flask and FastAPI for data integration, implementing proper security measures and caching strategies that improved data accessibility across systems by 50% while maintaining sub-second response times
- Established detailed technical documentation for data processes, pipelines, and architecture using Confluence and Azure DevOps Wiki, reducing onboarding time for new team members by 40% and improving knowledge sharing across teams
- Actively participated in Agile ceremonies and contributed to sprint planning and retrospectives, maintaining 95% sprint completion rate while ensuring high-quality deliverables and continuous improvement of team processes

CERTIFICATIONS

AWS Certified Data Engineer - Associate